D. Black and White Stripe

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: implementation, two pointers | https://codeforces.com/problemset/problem/1690/D

You have a stripe of checkered paper of length n. Each cell is either white or black.

What is the minimum number of cells that must be recolored from white to black in order to have a segment of k consecutive black cells on the stripe?

If the input data is such that a segment of k consecutive black cells already exists, then print `0`.

Input

The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

Next, descriptions of t test cases follow.

The first line of the input contains two integers n and k ($1 \le k \le n \le 2 \cdot 10^5$). The second line consists of the letters 'W' (white) and 'B' (black). The line length is n.

It is guaranteed that the sum of values n does not exceed $2 \cdot 10^5$.

Output

For each of t test cases print an integer — the minimum number of cells that need to be repainted from white to black in order to have a segment of k consecutive black cells.

Example

Input
4 5 3 BBWBW 5 5 BBWBW 5 1 BBWBW 1 1 W

Output
1 2 0 1

Note

In the first test case, s="BBWBW" and k=3. It is enough to recolor s_3 and get s="BBBBW". This string contains a segment of length k=3 consisting of the letters 'B'.

In the second test case of the example s="BBWBW" and k=5. It is enough to recolor s_3 and s_5 and get s="BBBBB". This string contains a segment of length k=5 consisting of the letters 'B'.

In the third test case of the example s="BBWBW" and k=1. The string s already contains a segment of length k=1 consisting of the letters 'B'.